

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1  
library(here)
```

```
## here() starts at /home/guest/project/EDE_Fall2025
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr   1.5.1  
## v ggplot2    3.5.1      v tibble    3.2.1  
## v lubridate  1.9.3      v tidyr     1.3.1  
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(ggthemes)
library(lubridate)
here()
```

```
## [1] "/home/guest/project/EDE_Fall2025"
```

```
NTL.raw <- read.csv(here('Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv'), stringsAsFactors = TRUE)
```

```
#2
my_theme <- theme_bw() +
  theme(
    plot.background = element_rect(fill = "azure"),
    panel.background = element_rect(fill = "linen"),
    legend.background = element_rect(fill = "antiquewhite"),

    # Modifying legend.position
    legend.position = 'top',

  )
theme_set(my_theme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

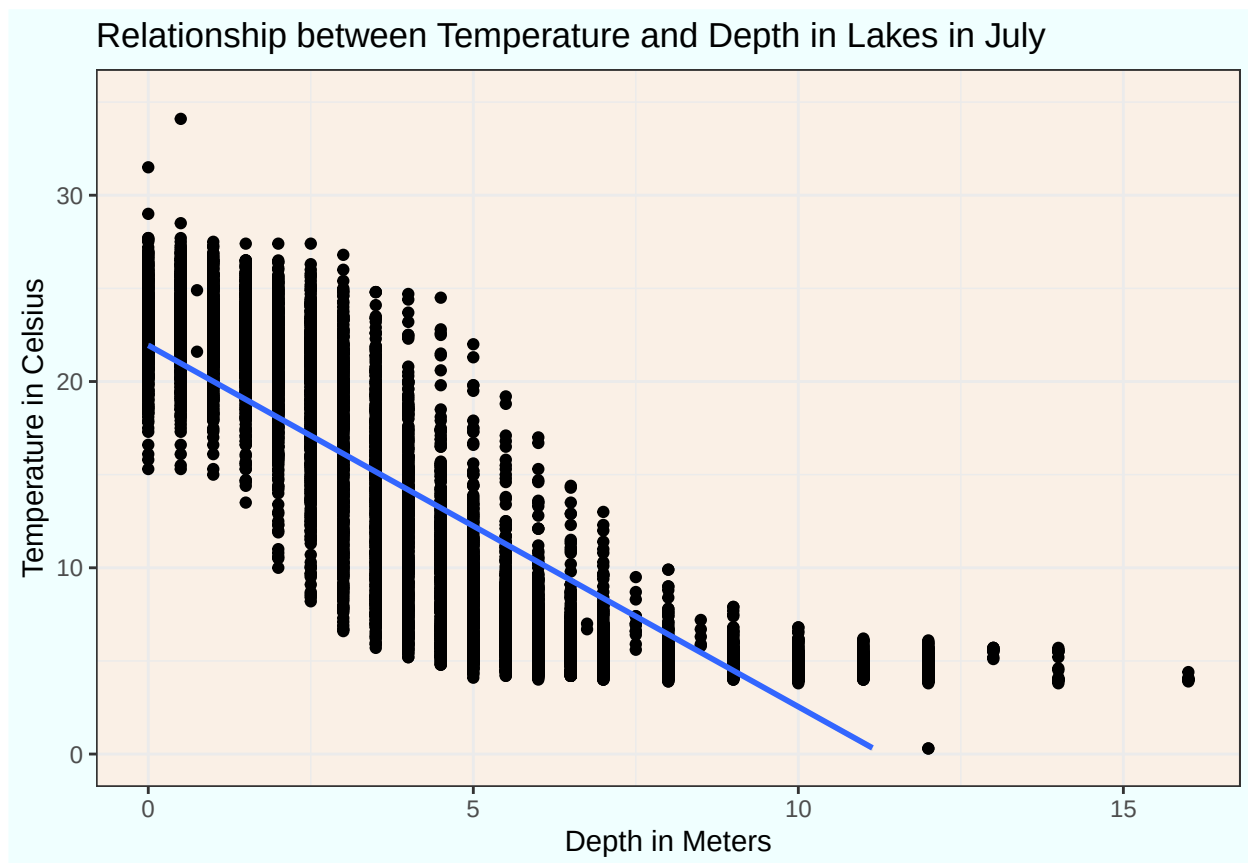
3. State the null and alternative hypotheses for this question: > Answer: H0: Lake temperature in July does not change with depth across all lakes Ha: Lake temperature in July changes with depth across all lakes
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
JulyLakes <- NTL.raw %>%
  filter(daynum >= 182 & daynum <= 213) %>%
  select("lakename", "year4", "daynum", "depth", "temperature_C") %>%
  drop_na()
```

```
#5
temp.depth.plot <- ggplot(data=JulyLakes, aes(x=depth, y=temperature_C))+
  geom_point()+
  geom_smooth(method = "lm")+
  my_theme+
  labs(x="Depth in Meters",
       y="Temperature in Celsius",
       title = "Relationship between Temperature and Depth in Lakes in July",
       legend)+
  ylim(0,35)
temp.depth.plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: Water gets warmer closer to the surface. There is a strong linearity in the data, suggesting a constant relationship between depth and temperature

7. Perform a linear regression to test the relationship and display the results.

```
#7
summary(lm(data=JulyLakes, depth ~ temperature_C ))

##
## Call:
## lm(formula = depth ~ temperature_C, data = JulyLakes)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.0778 -1.1159 -0.2393  0.9663  8.0868
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.584075   0.033561   285.6  <2e-16 ***
## temperature_C -0.379735   0.002272  -167.1  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.703 on 9974 degrees of freedom
## Multiple R-squared:  0.7368, Adjusted R-squared:  0.7368
## F-statistic: 2.793e+04 on 1 and 9974 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The linear regression shows that every additional meter of depth correlates to -0.38 degrees Celsius. This is very statistically significant, with a P-value far below 0.001. The variability in depth accounts for approximately 74% of the variability in Temperature. This is supported by a degrees of freedom just shy of 10,000 at 9,974 degrees.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
LakeAIC <- lm(data = JulyLakes, temperature_C ~ depth + year4 + daynum)

step(LakeAIC)
```

```
## Start: AIC=26781.56
## temperature_C ~ depth + year4 + daynum
##
##           Df Sum of Sq    RSS   AIC
## <none>                 146054 26782
## - year4    1         154 146209 26790
## - daynum   1        1582 147636 26887
## - depth    1       414049 560103 40189

##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = JulyLakes)
##
## Coefficients:
## (Intercept)      depth      year4      daynum
##   -14.33180    -1.94112     0.01386     0.04337

#10
summary(lm(formula = temperature_C ~ depth + year4 + daynum, data = JulyLakes))

##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = JulyLakes)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.669 -3.014  0.091  2.977 13.606
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -14.331802   8.582522  -1.670  0.09497 .
## depth       -1.941121   0.011545 -168.135 < 2e-16 ***
## year4        0.013861   0.004274   3.243  0.00119 **
## daynum       0.043368   0.004173  10.393 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.827 on 9972 degrees of freedom
## Multiple R-squared:  0.7399, Adjusted R-squared:  0.7398
## F-statistic: 9457 on 3 and 9972 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: AIC suggested keeping all the explanatory variables, depth, year4 and daynum. this model is statistically a very slight improvement in terms of our adjusted R-squared, increasing the value by a few hundreths, at the expense of 2 degrees of freedom. However, in terms of practical significant, I feel this makes communicating the essential relationships at play more challenging, which is a major drawback.

Analysis of Variance

2. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).¹

#12

```
JulyLake.aov <- aov(data=JulyLakes, temperature_C~lakenname)
summary(JulyLake.aov)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakenname      8  22407   2800.9    51.78 <2e-16 ***
## Residuals    9967 539167    54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
JulyLake.LM <- lm(data=JulyLakes, temperature_C~lakenname)
summary(JulyLake.LM)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakenname, data = JulyLakes)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.757  -6.579  -2.680   7.686  23.813
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.175 < 2e-16 ***
## lakennameCrampton Lake    -2.1851     0.7558  -2.891 0.003847 **
## lakennameEast Long Lake   -7.3795     0.6909 -10.681 < 2e-16 ***
## lakennameHummingbird Lake -6.5875     0.9287  -7.094 1.40e-12 ***
## lakennamePaul Lake        -3.8522     0.6655  -5.788 7.34e-09 ***
## lakennamePeter Lake       -4.3559     0.6641  -6.559 5.70e-11 ***
## lakennameTuesday Lake    -6.6094     0.6762  -9.775 < 2e-16 ***
## lakennameWard Lake        -3.2078     0.9428  -3.402 0.000671 ***
## lakennameWest Long Lake   -6.0878     0.6868  -8.864 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9967 degrees of freedom
## Multiple R-squared:  0.0399, Adjusted R-squared:  0.03913
## F-statistic: 51.78 on 8 and 9967 DF, p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

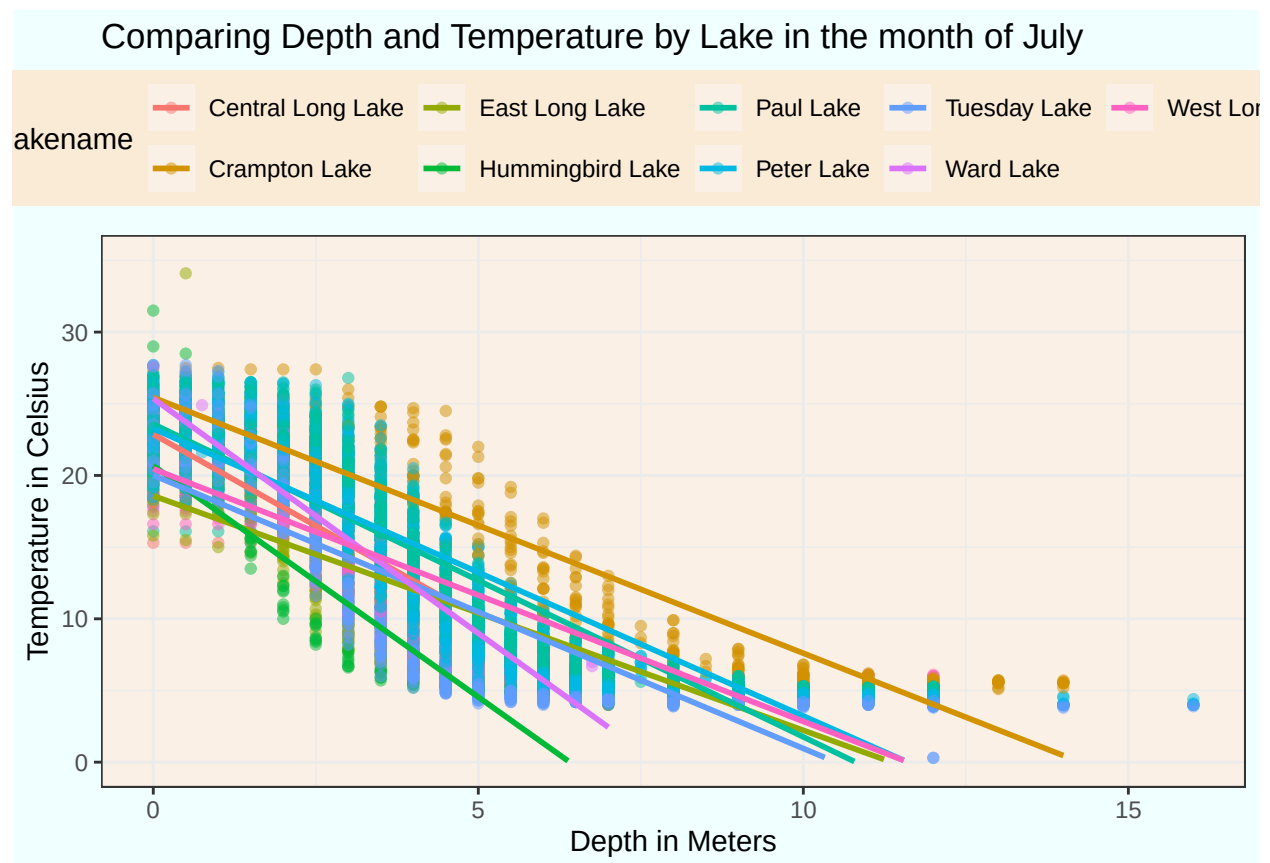
Answer: There is a statistically significant difference in mean temperatures among the lakes. The two tests agree that the estimated mean temperature is 12.7 degrees celsius, with a standard error of 0.075. The P value is very very far below alpha of 0.001. however, only 3.9% of the variability in temperature is explained by lake.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
JulyLakes.Plot2 <- ggplot(data=JulyLakes, aes(x=depth, y = temperature_C, color = lakename))+
  geom_point(alpha = 0.5)+
  geom_smooth(method="lm", se= FALSE)+
  ylim(0, 35)+
  my_theme+
  labs(x="Depth in Meters", y="Temperature in Celsius",
       title="Comparing Depth and Temperature by Lake in the month of July")
JulyLakes.Plot2
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
TukeyHSD(JulyLake.aov)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = JulyLakes)
##
## $lakename
##
```

	diff	lwr	upr	p adj
## Crampton Lake-Central Long Lake	-2.18508757	-4.5298903	0.1597151	0.0908865
## East Long Lake-Central Long Lake	-7.37946293	-9.5229855	-5.2359403	0.0000000
## Hummingbird Lake-Central Long Lake	-6.58754446	-9.4686472	-3.7064417	0.0000000
## Paul Lake-Central Long Lake	-3.85215061	-5.9169602	-1.7873410	0.0000003
## Peter Lake-Central Long Lake	-4.35586017	-6.4162973	-2.2954230	0.0000000
## Tuesday Lake-Central Long Lake	-6.60943694	-8.7071991	-4.5116748	0.0000000
## Ward Lake-Central Long Lake	-3.20778556	-6.1328832	-0.2826879	0.0193258
## West Long Lake-Central Long Lake	-6.08780244	-8.2184701	-3.9571348	0.0000000
## East Long Lake-Crampton Lake	-5.19437536	-6.5934427	-3.7953080	0.0000000
## Hummingbird Lake-Crampton Lake	-4.40245689	-6.7822580	-2.0226558	0.0000004
## Paul Lake-Crampton Lake	-1.66706304	-2.9422645	-0.3918615	0.0016569
## Peter Lake-Crampton Lake	-2.17077261	-3.4388820	-0.9026632	0.0000040
## Tuesday Lake-Crampton Lake	-4.42434937	-5.7522448	-3.0964539	0.0000000
## Ward Lake-Crampton Lake	-1.02269799	-3.4555762	1.4101802	0.9304479
## West Long Lake-Crampton Lake	-3.90271487	-5.2820063	-2.5234234	0.0000000
## Hummingbird Lake-East Long Lake	0.79191847	-1.3898338	2.9736707	0.9704234
## Paul Lake-East Long Lake	3.52731232	2.6771780	4.3774466	0.0000000
## Peter Lake-East Long Lake	3.02360276	2.1841440	3.8630615	0.0000000
## Tuesday Lake-East Long Lake	0.77002599	-0.1572797	1.6973317	0.1966860
## Ward Lake-East Long Lake	4.17167737	1.9321494	6.4112054	0.0000003
## West Long Lake-East Long Lake	1.29166049	0.2921434	2.2911776	0.0020066
## Paul Lake-Hummingbird Lake	2.73539385	0.6309242	4.8398635	0.0018237
## Peter Lake-Hummingbird Lake	2.23168429	0.1315045	4.3318640	0.0273701
## Tuesday Lake-Hummingbird Lake	-0.02189248	-2.1587031	2.1149182	1.0000000
## Ward Lake-Hummingbird Lake	3.37975890	0.4265319	6.3329859	0.0115771
## West Long Lake-Hummingbird Lake	0.49974202	-1.6693818	2.6688659	0.9986055
## Peter Lake-Paul Lake	-0.50370956	-1.1148073	0.1073882	0.2050877
## Tuesday Lake-Paul Lake	-2.75728633	-3.4843663	-2.0302063	0.0000000
## Ward Lake-Paul Lake	0.64436505	-1.5199444	2.8086745	0.9916947
## West Long Lake-Paul Lake	-2.23565183	-3.0528323	-1.4184714	0.0000000
## Tuesday Lake-Peter Lake	-2.25357677	-2.9681451	-1.5390084	0.0000000
## Ward Lake-Peter Lake	1.14807461	-1.0120638	3.3082130	0.7775157
## West Long Lake-Peter Lake	-1.73194227	-2.5380108	-0.9258737	0.0000000
## Ward Lake-Tuesday Lake	3.40165138	1.2058821	5.5974206	0.0000547
## West Long Lake-Tuesday Lake	0.52163450	-0.3755563	1.4188253	0.6798006
## West Long Lake-Ward Lake	-2.88001688	-5.1072441	-0.6527896	0.0019861

```
JulyLake.aovHSD <- HSD.test(JulyLake.aov, "lakename", group = TRUE)
JulyLake.aovHSD
```

```
## $statistics
## MSerror Df Mean CV
## 54.09519 9967 12.72153 57.81491
##
## $parameters
## test name.t ntr StudentizedRange alpha
## Tukey lakename 9 4.38748 0.05
```

```
##
## $means
##           temperature_C      std      r      se Min  Max  Q25  Q50
## Central Long Lake      17.66641 4.196292  128 0.6500913 8.9 26.8 14.40 18.40
## Crampton Lake          15.48132 7.347999  364 0.3855038 5.0 27.5  7.50 17.05
## East Long Lake         10.28694 6.765204  988 0.2339919 4.2 34.1  5.00  6.50
## Hummingbird Lake       11.07886 7.055590  123 0.6631729 4.0 31.5  5.25  7.70
## Paul Lake              13.81426 7.296928 2660 0.1426062 4.7 27.7  6.50 12.40
## Peter Lake             13.31055 7.671199 2930 0.1358769 4.0 27.0  5.60 11.40
## Tuesday Lake          11.05697 7.695476 1564 0.1859777 0.3 27.7  4.40  6.80
## Ward Lake              14.45862 7.409079  116 0.6828893 5.7 27.6  7.20 12.55
## West Long Lake         11.57860 6.952501 1103 0.2214581 4.0 25.7  5.40  8.00
##
##           Q75
## Central Long Lake 21.000
## Crampton Lake    22.400
## East Long Lake   16.025
## Hummingbird Lake 16.400
## Paul Lake        21.400
## Peter Lake        21.500
## Tuesday Lake     19.400
## Ward Lake        23.200
## West Long Lake   18.700
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.48132     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31055      c
## West Long Lake         11.57860      d
## Hummingbird Lake       11.07886     de
## Tuesday Lake          11.05697     de
## East Long Lake         10.28694      e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Peter Lake has statistically similar mean temperature as Paul and Ward lakes. No lake has a mean temperature that is statistically distinct from all other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: a Paired t-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does

the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

```
CramptonWardLakes <- JulyLakes %>%  
  filter(lakename == "Crampton Lake" | lakename == "Ward Lake")  
  
CramptonWard.ttest <- t.test(data=CramptonWardLakes, temperature_C ~ lakename)  
CramptonWard.ttest
```

```
##  
## Welch Two Sample t-test  
##  
## data: temperature_C by lakename  
## t = 1.2972, df = 192.4, p-value = 0.1961  
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0  
## 95 percent confidence interval:  
## -0.5323014 2.5776973  
## sample estimates:  
## mean in group Crampton Lake mean in group Ward Lake  
## 15.48132 14.45862
```

Answer: Crampton and Ward lakes are not different enough in their means to reject the null hypothesis given a p-value of 0.19. This supports the earlier answer that no lake has a mean that is distinct from all other lakes.